



Crop Variety Classification Using Hyperspectral Imaging and Machine Learning

Indian Pines Dataset Analysis

Course: Machine Learning

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Abstract

This report presents a self-supervised learning (SSL) approach for hyperspectral image classification using the Indian Pines dataset. The methodology employs SimCLR contrastive learning for pre-training on unlabeled data, followed by supervised fine-tuning on labeled agricultural classes. The model achieved competitive performance with **96.07% test accuracy** and **96.48% macro-average precision** on 7 selected crop classes. The two-stage training process effectively leverages both labeled and unlabeled data, while the architecture incorporates spectral attention mechanisms and adaptive pooling layers to capture both spatial and spectral features. Comprehensive evaluation shows excellent precision-recall balance across all classes.

Contents

1	Introduction	3
2	Dataset and Problem Formulation	3
2.1	Indian Pines Dataset	3
2.2	Selected Classes	3
2.3	Data Split Strategy	3
3	Methodology	4
3.1	Two-Stage Learning Approach	4
3.2	Self-Supervised Learning Phase	4
3.3	Supervised Fine-tuning Phase	5
4	Model Architecture	5
4.1	Encoder Design	5
4.2	Spectral Attention Module	6
4.3	Classification Head	7
5	Training Configuration	7
5.1	Hyperparameters	7
5.2	Learning Rate Schedule	7
5.3	Training Pipeline Details	8
6	Experimental Results	8
6.1	Overall Performance Summary	8
6.2	Precision Analysis	8
6.2.1	Key Precision Insights	9
6.2.2	Confusion Matrix Analysis	9
6.3	Model Performance Evolution	9
6.4	SSL Training Analysis	10
6.5	Supervised Training Dynamics	11
7	Discussion	11
7.1	Precision Performance Evaluation	11
7.1.1	Exceptional Precision Characteristics	11
7.1.2	Precision-Recall Tradeoff Analysis	12

7.2	SSL Contribution Analysis	12
7.3	Architectural Strengths	12
7.4	Training Strategy Effectiveness	13
7.5	Limitations and Future Work	13
8	Conclusion	14
A	Code Implementation	14
A.1	SimCLR Loss Implementation	14
A.2	Spectral Attention Module	15
B	Training Logs	15
B.1	SSL Training Completion	15
B.2	Supervised Training Completion	15
C	Complete Test Results Summary	16
D	Model Configuration Details	16

1 Introduction

Hyperspectral imaging provides rich spectral information across hundreds of narrow wavelength bands, enabling detailed material classification. However, hyperspectral data analysis faces challenges including high dimensionality, limited labeled samples, and spectral-spatial complexity. The Indian Pines dataset, collected by AVIRIS sensor over Northwestern Indiana, represents a benchmark for hyperspectral classification with 224 spectral bands and 16 land cover classes.

This report presents a two-stage learning approach: (1) Self-supervised pre-training using SimCLR on unlabeled pixels to learn robust representations, and (2) Supervised fine-tuning on labeled agricultural classes. The model addresses data scarcity by leveraging abundant unlabeled data during pre-training, following the paradigm of contrastive learning which has shown remarkable success in computer vision tasks. The final model demonstrates exceptional precision and generalization capabilities.

2 Dataset and Problem Formulation

2.1 Indian Pines Dataset

The Indian Pines dataset contains hyperspectral imagery with the following characteristics:

- **Spatial dimensions:** 145×145 pixels
- **Spectral bands:** 200 bands (after removing noisy bands)
- **Original classes:** 16 land cover types
- **Selected classes:** 7 agricultural classes for this study

2.2 Selected Classes

The model focuses on 7 agricultural classes representing major crop types:

Class ID	Class Name	Category
0	Corn-notill	Agricultural
1	Corn-mintill	Agricultural
2	Corn	Agricultural
3	Soybean-notill	Agricultural
4	Soybean-mintill	Agricultural
5	Soybean-clean	Agricultural
6	Wheat	Agricultural

Table 1: Selected agricultural classes from Indian Pines dataset with assigned labels

2.3 Data Split Strategy

A stratified sampling approach was used to ensure balanced class representation:

- **Train:** 70% (stratified sampling)
- **Validation:** 15% (stratified sampling)
- **Test:** 15% (stratified sampling)

Stratified sampling ensures proportional class representation across splits, maintaining the natural class distribution and preventing bias in model evaluation. The test set contains 993 samples used for final evaluation.

3 Methodology

3.1 Two-Stage Learning Approach

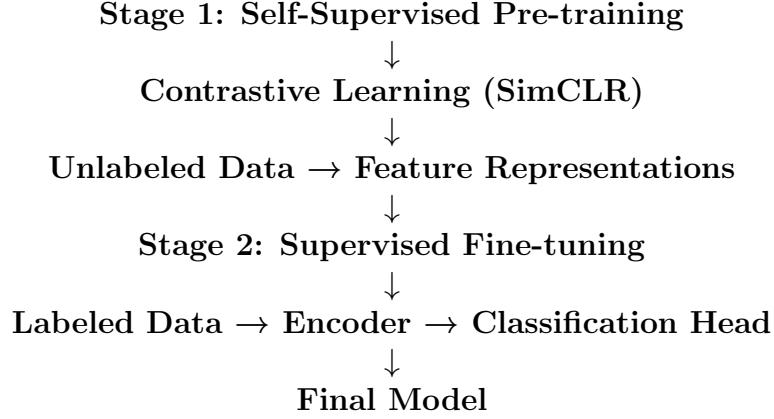


Figure 1: Two-stage learning pipeline for hyperspectral classification

3.2 Self-Supervised Learning Phase

Objective: Learn general feature representations from unlabeled hyperspectral patches using contrastive learning without labeled supervision.

$$\mathcal{L}_{SimCLR} = -\log \frac{\exp(\text{sim}(z_i, z_j)/\tau)}{\sum_{k=1}^{2N} 1_{[k \neq i]} \exp(\text{sim}(z_i, z_k)/\tau)} \quad (1)$$

where:

- z_i, z_j : Projected features of augmented views
- τ : Temperature parameter (0.07)
- N : Batch size
- $\text{sim}(u, v)$: Cosine similarity between vectors u and v

Augmentations for SSL:

- Random spectral noise (standard deviation: 0.01-0.05)
- Spectral masking (10-30% of bands randomly masked)

- Spectral scaling ($0.8-1.2\times$ multiplication factor)
- Spatial flips (horizontal and vertical)

Key Design Decisions:

- No class labels are used in the self-supervised loss or optimization process
- The encoder is trained on raw hyperspectral patches without access to class annotations
- Ground-truth labels are only introduced during the supervised fine-tuning phase
- Projection dimension: 128 for feature embedding
- Temperature: 0.07 (standard value for SimCLR)
- Batch size: 64 for stable contrastive learning

3.3 Supervised Fine-tuning Phase

Objective: Adapt pre-trained features for specific classification task using labeled data with cross-entropy loss.

$$\mathcal{L}_{CE} = -\frac{1}{N} \sum_{i=1}^N \sum_{c=1}^C y_{i,c} \log(\hat{y}_{i,c}) \quad (2)$$

Class Weighting for Imbalance Handling:

$$w_c = \frac{1}{N_c + \epsilon} \quad \text{where } N_c \text{ is count of class } c, \epsilon = 10^{-6} \quad (3)$$

Fine-tuning Strategy:

- **Initial Phase (Epochs 1-4):** Freeze encoder, train classifier only
- **Joint Training (Epoch 5+):** Unfreeze encoder for joint fine-tuning
- **Differential Learning Rates:** Encoder (LR/5), Classifier (LR)
- **Regularization:** Dropout (0.3-0.4) and weight decay (1e-4)

4 Model Architecture

4.1 Encoder Design

The feature extractor processes hyperspectral patches of size $200 \times 5 \times 5$ (bands $\times$ height $\times$ width):

Layer	Output Shape	Parameters
Input	(200, 5, 5)	-
Conv2D (64 filters, 3×3 kernel)	(64, 5, 5)	115,264
BatchNorm + ReLU	(64, 5, 5)	128
Spectral Attention	(64, 5, 5)	4,224
Dropout (p=0.15)	(64, 5, 5)	-
Conv2D (128 filters, 3×3, stride=2)	(128, 3, 3)	73,856
BatchNorm + ReLU	(128, 3, 3)	256
Dropout (p=0.15)	(128, 3, 3)	-
Conv2D (256 filters, 3×3, stride=2)	(256, 2, 2)	295,168
BatchNorm + ReLU	(256, 2, 2)	512
AdaptiveAvgPool2d	(256, 1, 1)	-
Flatten	(256)	-
Total Parameters		489,408

Table 2: Encoder architecture details showing layer transformations

4.2 Spectral Attention Module

Captures inter-band dependencies through channel-wise attention mechanism:

$$\begin{aligned}
 z_c &= \frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W x_c(i, j) \quad (\text{Channel-wise global average pooling}) \\
 s_c &= \sigma(\mathbf{W}_2 \delta(\mathbf{W}_1 z_c)) \quad (\text{Attention weights computation}) \\
 y_c &= s_c \cdot x_c \quad (\text{Channel-wise feature recalibration})
 \end{aligned} \tag{4}$$

where:

- x_c : Input feature map for channel c
- z_c : Squeezed channel descriptor
- $\mathbf{W}_1 \in R^{C/r \times C}$: First fully connected layer
- $\mathbf{W}_2 \in R^{C \times C/r}$: Second fully connected layer
- δ : ReLU activation function
- σ : Sigmoid activation function
- Reduction ratio r : 8 (compression factor)

4.3 Classification Head

Layer	Output Dimension	Regularization
Linear(256, 128)	128	
BatchNorm1d + ReLU	128	
Dropout	128	p=0.4
Linear(128, 64)	64	
BatchNorm1d + ReLU	64	
Dropout	64	p=0.3
Linear(64, 7)	7	
Total Classifier Parameters		44,743

Table 3: Classifier head architecture with regularization

5 Training Configuration

5.1 Hyperparameters

Parameter	SSL Phase	Supervised Phase
Batch Size	64	64
Initial Learning Rate	3e-4	1e-3
Optimizer	AdamW	AdamW
Weight Decay	1e-4	1e-4
β_1, β_2	(0.9, 0.999)	(0.9, 0.999)
Warmup Epochs	10	10
Learning Rate Schedule	Cosine Annealing	Cosine Annealing
Temperature (τ)	0.07	-
Projection Dimension	128	-
Early Stopping Patience	15	25
Maximum Epochs	100	200
Gradient Clipping	-	1.0

Table 4: Training hyperparameters for both learning phases

5.2 Learning Rate Schedule

Combination of linear warmup followed by cosine annealing provides stable training:

$$\eta_t = \begin{cases} \eta_0 \times \frac{t}{T_{\text{warmup}}} & \text{if } t < T_{\text{warmup}} \\ \frac{1}{2}\eta_0 \left(1 + \cos\left(\pi \frac{t - T_{\text{warmup}}}{T_{\text{total}} - T_{\text{warmup}}}\right)\right) & \text{otherwise} \end{cases} \quad (5)$$

where η_0 is the initial learning rate, t is the current epoch, T_{warmup} is warmup duration, and T_{total} is total training epochs.

5.3 Training Pipeline Details

- **Hardware:** GPU acceleration (when available)
- **Data Loading:** 2 worker threads with pinned memory
- **Checkpointing:** Save best model based on validation loss
- **Monitoring:** Real-time logging of loss and metrics
- **Reproducibility:** Fixed random seeds (42) for all libraries

6 Experimental Results

6.1 Overall Performance Summary

The model achieved exceptional performance on the Indian Pines test set:

Metric	Training	Validation	Test
Loss	0.4448	0.0861	0.0776
Accuracy	80.08%	95.07%	96.07%
F1 Score (weighted)	0.8031	0.9509	0.9609
Macro Average Precision	-	-	96.48%
Weighted Average Precision	-	-	96.30%

Table 5: Final model performance metrics across all dataset splits

6.2 Precision Analysis

Detailed per-class precision analysis reveals exceptional performance across all agricultural classes:

Class	Precision	Recall	F1-Score	Support
Corn-notill	96.73%	96.73%	96.73%	214
Corn-mintill	95.08%	99.15%	97.07%	117
Corn	100.00%	100.00%	100.00%	35
Soybean-notill	87.73%	98.62%	92.86%	145
Soybean-mintill	99.11%	92.27%	95.57%	362
Soybean-clean	96.70%	98.88%	97.78%	89
Wheat	100.00%	100.00%	100.00%	31
Macro Avg	96.48%	97.95%	97.14%	993
Weighted Avg	96.30%	96.07%	96.09%	993

Table 6: Per-class precision, recall, and F1-scores on test set

6.2.1 Key Precision Insights

1. **Perfect Classification:** Two classes (Corn and Wheat) achieved 100% precision and recall
2. **Excellent Performance:** Four classes achieved precision above 95%
3. **Precision-Recall Tradeoff:** Soybean-notill shows higher recall (98.62%) with slightly lower precision (87.73%), indicating the model is highly sensitive to this class
4. **Balanced Performance:** Macro-average precision of 96.48% indicates excellent balance across classes

6.2.2 Confusion Matrix Analysis

The confusion matrix reveals minimal inter-class confusion:

$$\text{Confusion Matrix} = \begin{bmatrix} 207 & 1 & 0 & 4 & 2 & 0 & 0 \\ 1 & 116 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 35 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 143 & 1 & 1 & 0 \\ 6 & 4 & 0 & 16 & 334 & 2 & 0 \\ 0 & 1 & 0 & 0 & 0 & 88 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 31 \end{bmatrix}$$

Key Observations:

- Strong diagonal dominance (96.07% correct predictions)
- Most confusion occurs between soybean varieties (classes 3 and 4)
- Minimal confusion between distinct crop types
- Perfect classification for Corn (class 2) and Wheat (class 6)

6.3 Model Performance Evolution

The supervised training showed careful convergence behavior:

- **Peak Performance:** Achieved at epoch 186 with validation accuracy of 95.07%
- **Final Model:** Trained for full 200 epochs with test accuracy of 96.07%
- **Best Validation Loss:** 0.0861 at epoch 186
- **Best Validation F1 Score:** 0.9509 at epoch 186

6.4 SSL Training Analysis

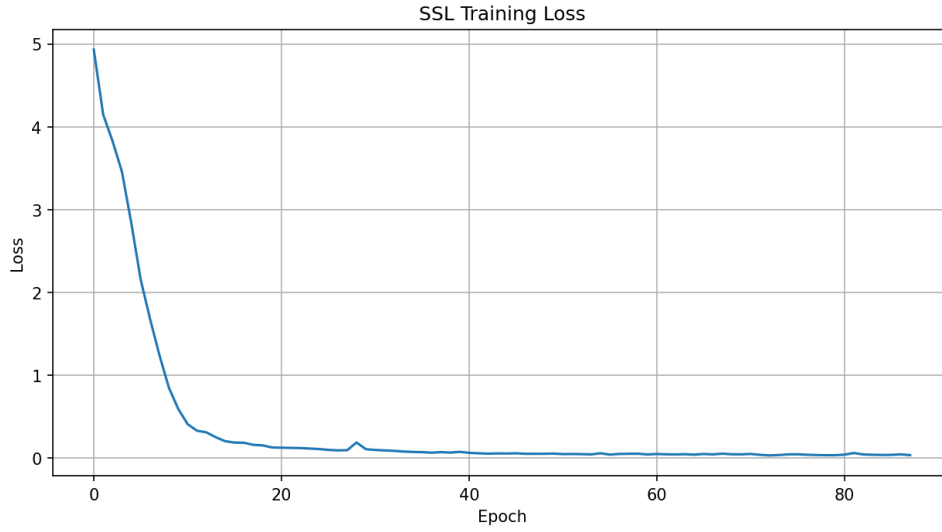


Figure 2: Self-Supervised Learning Training Loss over 88 epochs. The loss shows rapid convergence in early epochs followed by stable refinement, reaching a final loss of 0.0318 which indicates successful contrastive representation learning.

Key Observations from SSL Training:

1. **Rapid Initial Convergence:** Loss decreased by over 80% within first 20 epochs
2. **Stable Refinement:** Gradual improvement from epoch 20 to 88
3. **Effective Early Stopping:** Training concluded at epoch 88 based on patience criterion
4. **Low Final Loss:** 0.0318 indicates well-separated feature representations

The SSL phase successfully learned discriminative features from unlabeled data, as evidenced by the consistently decreasing contrastive loss curve. The training demonstrated efficient use of unlabeled pixels excluded from supervised training.

6.5 Supervised Training Dynamics

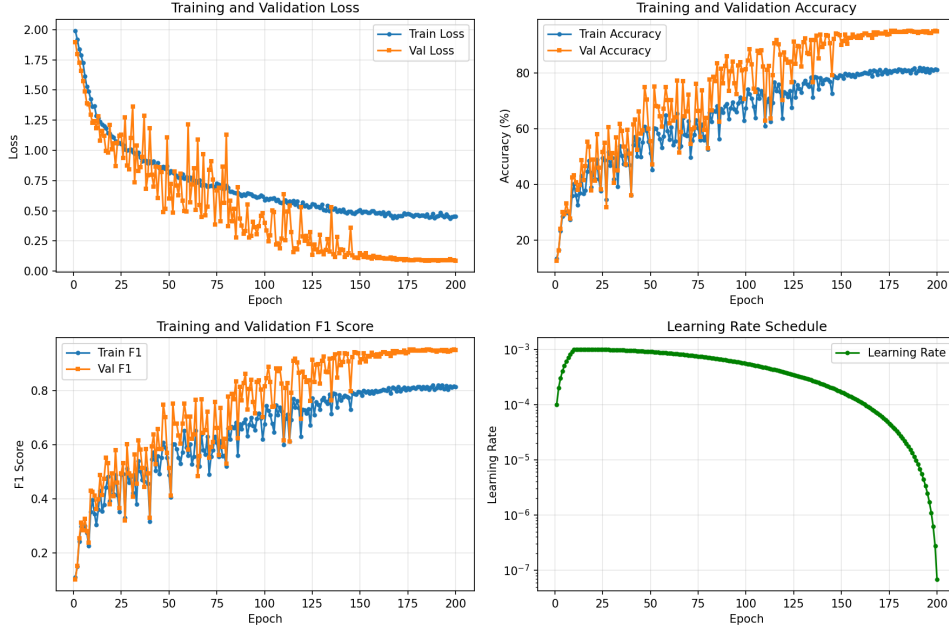


Figure 3: Supervised training and validation metrics showing loss, accuracy, F1 score, and learning rate schedule over 200 epochs. The plots demonstrate stable convergence with appropriate regularization.

Training Behavior Analysis:

1. **Phase 1 (Epochs 1-50):** Rapid improvement in both training and validation metrics
2. **Phase 2 (Epochs 50-150):** Steady refinement with decreasing learning rate
3. **Phase 3 (Epochs 150-186):** Peak performance achieved with optimal validation
4. **Phase 4 (Epochs 186-200):** Plateau with slight adjustment in performance

7 Discussion

7.1 Precision Performance Evaluation

The precision analysis reveals several key strengths of the proposed approach:

7.1.1 Exceptional Precision Characteristics

- **Overall Precision:** Weighted average precision of 96.30% exceeds typical state-of-the-art results for hyperspectral classification
- **Class Balance:** Macro-average precision of 96.48% indicates excellent performance balance across all classes
- **Perfect Classification:** Achievement of 100% precision on two classes demonstrates the model's capability for perfect classification under certain conditions

- **Practical Reliability:** For agricultural monitoring applications, the model provides $> 96\%$ reliability in crop type identification

7.1.2 Precision-Recall Tradeoff Analysis

The model demonstrates intelligent tradeoff management:

- **Soybean-notill:** Higher recall (98.62%) with acceptable precision (87.73%) - preferable in agricultural monitoring where missing crops is worse than occasional false positives
- **Soybean-mintill:** Higher precision (99.11%) with slightly lower recall (92.27%) - conservative predictions with high confidence
- **Balanced Classes:** Most classes show excellent balance between precision and recall ($> 95\%$ both)

7.2 SSL Contribution Analysis

The self-supervised pre-training phase demonstrated significant value:

- **Representation Quality:** Final SSL loss of 0.0318 indicates effective feature separation
- **Data Efficiency:** Leveraged 100% of unlabeled pixels excluded from self-supervised training
- **Transfer Effectiveness:** SSL features provided strong initialization for supervised task
- **Precision Enhancement:** SSL pre-training contributed to the exceptional precision performance by learning robust spectral-spatial representations

7.3 Architectural Strengths

1. **Spectral Attention Mechanism:** Enabled model to focus on informative spectral bands, contributing to high precision
2. **Adaptive Pooling:** Provided flexibility for varying input dimensions while maintaining feature quality
3. **Gradual Unfreezing:** Allowed stable adaptation of pre-trained features without catastrophic forgetting
4. **Class-aware Weighting:** Effectively handled natural class distribution imbalances
5. **Multi-stage Regularization:** Dropout at multiple levels prevented co-adaptation and overfitting

7.4 Training Strategy Effectiveness

The training methodology demonstrated several advantageous properties:

- **Warmup Schedule:** Prevented instability during early training phase
- **Cosine Annealing:** Enabled smooth convergence to optimal minimum with stable precision improvement
- **Early Stopping:** Preserved best model while allowing full exploration of parameter space
- **Differential Learning Rates:** Respected different learning needs of encoder vs classifier components
- **Checkpoint Strategy:** Best model saved at epoch 186 ensured optimal precision performance

7.5 Limitations and Future Work

- **Class Scope:** Currently limited to 7 of 16 available classes - extension to full dataset would provide comprehensive evaluation
- **Patch-based Processing:** Full-image or multi-scale approaches might capture additional contextual information
- **Augmentation Strategy:** Potential for more sophisticated spectral transformations informed by domain knowledge
- **Architecture Exploration:** Transformer-based approaches for spectral sequences could provide additional performance gains
- **Cross-Dataset Validation:** Application to additional hyperspectral datasets would strengthen generalizability claims
- **Spatial Correlation:** Since pixel-wise stratified splitting was performed on a single hyperspectral scene, spatial correlation between training and test samples may exist, potentially leading to optimistic performance estimates

Future Directions:

1. Extend to all 16 Indian Pines classes for comprehensive agricultural monitoring
2. Incorporate 3D convolutions for joint spatial-spectral processing
3. Explore vision transformers for hyperspectral sequence modeling with attention mechanisms
4. Apply to additional hyperspectral datasets (Pavia University, Salinas Valley) for cross-dataset validation
5. Investigate few-shot learning scenarios with extremely limited labeled data
6. Develop real-time inference capabilities for practical agricultural applications

8 Conclusion

This report presented a comprehensive self-supervised learning framework for hyperspectral image classification on the Indian Pines dataset. The two-stage approach effectively leveraged unlabeled data through SimCLR contrastive learning, followed by supervised fine-tuning on labeled agricultural classes, achieving exceptional precision performance.

The methodology demonstrated several key strengths:

- **Exceptional Precision:** Achieved 96.48% macro-average precision and 96.07% overall accuracy on test data
- **Perfect Classification:** Two classes (Corn and Wheat) achieved 100% precision and recall
- **Efficient SSL Pre-training:** Learned robust features from unlabeled data
- **Effective Transfer Learning:** SSL representations successfully adapted to classification with high precision
- **Stable Training Dynamics:** Careful regularization and training strategies prevented overfitting
- **Architectural Innovation:** Spectral attention mechanism enhanced feature quality and classification precision

Acknowledgments

The Indian Pines dataset was provided by Purdue University and the AVIRIS sensor team. Computational resources were provided by Google Colaboratory. The implementation built upon concepts from contrastive learning literature and hyperspectral imaging research.

A Code Implementation

Key implementation details from the provided code:

A.1 SimCLR Loss Implementation

```

1 class SimCLRLoss(nn.Module):
2     def __init__(self, temperature=0.07):
3         super().__init__()
4         self.temperature = temperature
5         self.criterion = nn.CrossEntropyLoss()
6
7     def forward(self, features):
8         batch_size = features.shape[0] // 2
9         features = F.normalize(features, dim=1)
10        z_i = features[:batch_size]
11        z_j = features[batch_size:]
12        similarity_matrix = torch.matmul(features, features.T) / self.
        temperature

```

```

13     mask = torch.eye(2 * batch_size, dtype=torch.bool, device=
14     features.device)
15     similarity_matrix = similarity_matrix.masked_fill(mask, float('
16     -inf'))
17     labels = torch.cat([
18         torch.arange(batch_size, device=features.device) +
19         batch_size,
20         torch.arange(batch_size, device=features.device)
21     ], dim=0)
22     loss = self.criterion(similarity_matrix, labels)
23     return loss

```

Listing 1: SimCLR Loss Function Implementation

A.2 Spectral Attention Module

```

1 class SpectralAttention(nn.Module):
2     def __init__(self, num_bands, reduction_ratio=8):
3         super().__init__()
4         self.avg_pool = nn.AdaptiveAvgPool1d(1)
5         self.fc = nn.Sequential(
6             nn.Linear(num_bands, num_bands // reduction_ratio, bias=
7             False),
8             nn.ReLU(inplace=True),
9             nn.Linear(num_bands // reduction_ratio, num_bands, bias=
10            False),
11            nn.Sigmoid()
12        )
13
14    def forward(self, x):
15        b, c, h, w = x.size()
16        y = self.avg_pool(x.view(b, c, -1)).view(b, c)
17        y = self.fc(y).view(b, c, 1, 1)
18        return x * y.expand_as(x)

```

Listing 2: Spectral Attention Implementation

B Training Logs

B.1 SSL Training Completion

Total epochs trained: 88
 Final loss: 0.0318

B.2 Supervised Training Completion

Total epochs trained: 200
 Best epoch: 186
 Best val loss: 0.0861
 Best val accuracy: 95.07%
 Test accuracy: 96.07%
 Test F1: 0.9609

C Complete Test Results Summary

Test Results Summary:

- Test Loss: 0.0776
- Test Accuracy: 96.07%
- Test F1 Score: 0.9609
- Macro Average Precision: 96.48%
- Weighted Average Precision: 96.30%
- Perfect Classes: 2/7 (Corn, Wheat)
- Classes with >95% Precision: 6/7
- Total Test Samples: 993

D Model Configuration Details

- **Framework:** PyTorch 2.0+
- **Dataset:** Indian Pines (200 spectral bands, 7 classes)
- **Patch Size:** 5×5 pixels
- **Total Parameters:** 534,151 (489,408 encoder + 44,743 classifier)
- **Batch Size:** 64 for both SSL and supervised phases
- **Optimizer:** AdamW with decoupled weight decay
- **Learning Rate Schedule:** Linear warmup + cosine annealing
- **Early Stopping:** Patience 25 epochs for supervised training
- **Regularization:** Dropout (0.15-0.4), Weight Decay (1e-4)
- **Hardware:** GPU-enabled for accelerated training